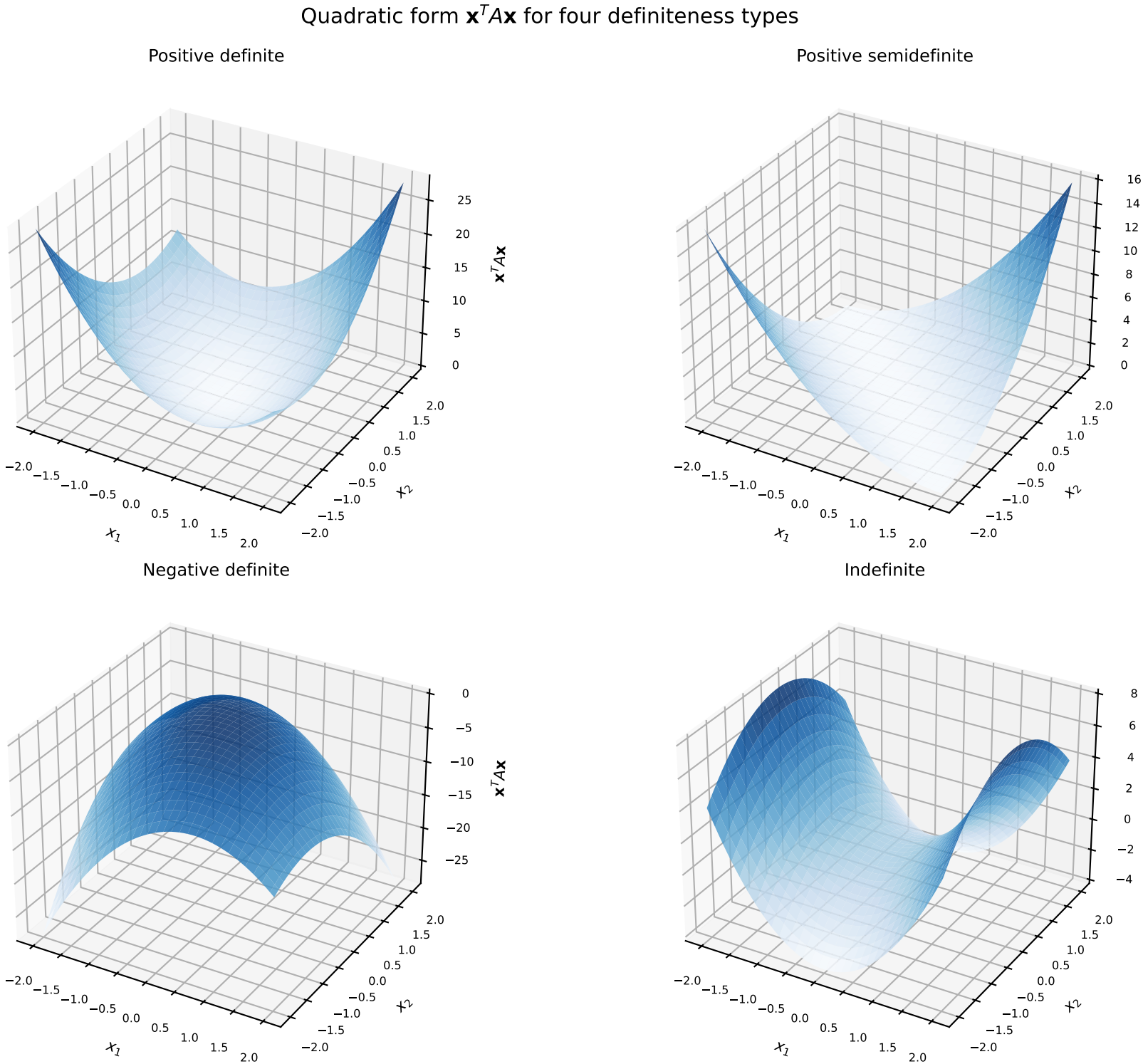
Quadratic form $\mathbf{x}^T A\mathbf{x}$ for four definiteness types
Positive definite
Positive semidefinite
Negative definite
Indefinite
$\mathbf{x}^T A\mathbf{x}$
x_1
x_2